ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 1 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording.

SECTION A

QUESTION 1

1.1 B ✓
1.2 D ✓
1.3 A ✓
1.4 C ✓
1.5 A ✓
1.6 B ✓
1.7 D ✓
1.8 C ✓
1.9 A ✓
1.10 B ✓

(10 × 1) [10]

QUESTION 2

2.1 True ✓
2.2 True ✓
2.3 False ✓
2.4 False ✓
2.5 True ✓

(5 × 1) [5]

QUESTION 3

3.1 teach pendant ✓
3.2 slicer ✓
3.3 latency ✓
3.4 LiDAR ✓
3.5 mesh network ✓

(5 × 1) [5]

TOTAL SECTION A: [20]

SECTION B

QUESTION 4

4.1 Electric: accurate, clean and easy to control ✓ / limited force for its size, needs gearboxes ✓. Hydraulic: very large forces, holds loads well ✓ / oil leaks, noisy, needs a power pack ✓. Pneumatic: fast, cheap, clean ✓ / air is compressible, so accurate positioning is difficult ✓. (6)

4.2 Point-to-point: the robot moves between taught points and the path between them is not controlled (e.g. spot welding, pick and place) ✓. Continuous path: the whole path is controlled, so the tool follows an exact line or curve (e.g. arc welding, painting, gluing) ✓. (2)

4.3 Any sensible examples ✓✓✓, e.g. 4.3.1 lights, alarms or geysers controlled from a phone; 4.3.2 soil-moisture sensors that switch irrigation on, livestock trackers; 4.3.3 wearable heart-rate monitors that send data to a doctor. (3)

4.4 Any ONE risk with its solution ✓✓, e.g. hackers taking control of devices or stealing data – use strong passwords, encryption, firmware updates and a separate network for IoT devices. (2)

[13]

QUESTION 5

5.1 In order ✓✓✓✓: design the finger in CAD; export it as an STL file; slice it (set layer height, infill, supports) to produce G-code; print it; remove supports and finish (sand, fit soft pads), then test the fit. (4)

5.2 A higher infill percentage makes the part stronger and heavier ✓ but takes longer and uses more filament ✓. Thinner layers give a smoother surface and finer detail ✓ but increase the printing time; thicker layers print faster with a rougher finish ✓. (4)

5.3 Printed parts are never exactly the designed size (plastic shrinks and spreads slightly) ✓; a small clearance makes sure holes and mating parts still fit together without forcing or being too loose ✓. (2)

5.4 Any THREE with properties ✓✓✓: PLA – easy to print, rigid, low temperature; ABS – tough, heat resistant, needs a heated bed / enclosure; PETG – strong, slightly flexible, food safe; TPU – flexible, rubbery (good for soft grip pads); nylon – very strong and wear resistant. (3)

[13]

QUESTION 6

6.1 Star: every device is connected to a central switch ✓✓; advantage – one faulty cable affects only one device / easy to add devices ✓. Bus: all devices share one main cable (backbone) ✓✓; advantage – uses little cable, cheap and simple ✓. (6)

6.2.1 The amount of data that can be sent over a network connection in a given time ✓, measured in bits per second (e.g. Mbps) ✓. (2)

6.2.2 A unique numerical address given to every device on a network ✓ so that data can be sent to and from the right device ✓. (2)

6.3 Any TWO fieldbus ✓✓: Profibus, Profinet, Modbus, DeviceNet, CANopen, EtherNet/IP, AS-i. Any TWO wireless ✓✓: Wi-Fi, Bluetooth, ZigBee, cellular (4G / 5G), LoRaWAN. (4)

6.4 A switch connects devices within the same local network and sends data to the right device using MAC addresses ✓; a router connects different networks (e.g. the plant network to the internet) and directs data using IP addresses ✓. (2)

[16]

QUESTION 7

7.1.1 Six axes ✓ (1)

7.1.2 Reach: the maximum distance from the robot's base to the wrist / tool that the arm can extend to ✓✓. Repeatability: how closely the robot returns to the same taught point again and again (e.g. ± 0,05 mm) ✓✓. (4)

7.2 $\mathrm{F} = \mathrm{m} \times \mathrm{g} = 12 \times 9{,}8$ ✓ $= 117{,}6\ \mathrm{N}$ ✓; $\tau = \mathrm{F} \times \mathrm{r}$ ✓ $= 117{,}6 \times 0{,}4$ ✓ $= 47{,}04\ \mathrm{N \cdot m}$ ✓ (5)

7.3 $\mathrm{W} = 5 \times 9{,}8 = 49\ \mathrm{N}$ ✓; $\mathrm{F} = \mathrm{W} \div (\mu \times \mathrm{n})$ ✓ $= 49 \div (0{,}5 \times 2)$ ✓ $= 49\ \mathrm{N\ per\ finger}$ ✓ (4)

7.4 Any THREE ✓✓✓: assembling small electronic parts; pick and place; packaging; screwing and inserting parts; dispensing glue; laboratory sample handling. (3)

[17]

QUESTION 8

8.1 With the right hand, the thumb points along +X, the index finger along +Y and the middle finger along +Z ✓; this gives a standard, consistent direction for all three axes of the robot's frame ✓. (2)

8.2 The analogue sensor gives a varying voltage that the microcontroller's ADC changes into a number (e.g. 0–1023) ✓✓; the program uses the value to decide the speed and sends a PWM signal through a motor driver, where a larger duty cycle gives a higher average voltage and speed ✓✓. (4)

8.3 A mechanical switch's contacts bounce, giving several quick on-off signals when it is pressed once ✓; debouncing (a short delay or a capacitor) makes the microcontroller read only one clean press ✓. (2)

8.4 The microcontroller can supply only a few milliamps at 5 V ✓; the relay lets this small signal switch the 230 V lamp safely, keeping the high-voltage circuit electrically isolated from the microcontroller ✓. (2)

8.5 The TCP is the working point at the tip of the tool (e.g. the centre between gripper fingers or the tip of a welding torch) ✓; all positions and movements are calculated for this point ✓, so if it is set wrongly the tool misses its target or collides ✓. (3)

[13]

QUESTION 9

9.1 Any TWO ✓✓: breadboard view; schematic (circuit diagram) view; PCB view. (2)

9.2 It shows the logic and order of the steps and decisions before coding ✓, so errors are found early and the program is easier to write and debug ✓. (2)

9.3 Any FOUR ✓✓✓✓: does it perform the required function; accuracy / repeatability; speed or cycle time; safety; reliability over many cycles; power consumption / battery life; cost; ease of use. (4)

[8]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE